Summary of Revisions

Trace Data Quality for LLM-Agent Observability (BigD260, transferred to AAI-BD 2026)

We thank the four reviewers. Their central concern was that the main results followed from how the schema projections were defined. We agree. In response we added a new experiment that measures what real instrumentation libraries emit, reframed the projection study as a specification-level analysis, and removed results that came from prior work or did not test the paper's question. Section and table numbers refer to the revised paper.

Main changes

1. **New experiment E1 (Sec. IV-C, V-C; Figs. 2–3; Tables IV, V, VIII, IX).** One orchestration-heavy agent with planning, memory, a guardrail and delegation runs 44 tickets, both as plain SDK code and as a LangGraph application. Each execution is recorded once and replayed identically under OpenInference, OpenLLMetry, the official OTel GenAI instrumentation, the Anthropic SDK's own spans and an agent-aware reference, then scored instance by instance against independent ground truth. The protocol was committed before any run; the one later addition is disclosed in Sec. IV-C.
2. **Projection study reframed (Sec. I, IV-B, V-A).** We state that its answerability follows from the mapping rules and use E1 to test it. We also found that our OTel GenAI anchor was stale: the cited commit was in the main semantic-conventions repository after the GenAI conventions had moved to their own repository, which already defined `invoke_agent`, `plan` and memory operations. The view is relabeled *GenAI model/tool*, and the paper says it understates the conventions.
3. **Removed or condensed.** The controlled fault-matrix results (former Tables XIV–XVI), the diagnostic-proxy table (XIII), the per-corpus table (VIII), the denominator table (VI) and the three τ-bench tables (IX–XI) are removed or condensed into text. Sections II-B (boundary) and VI-C (why BigData) are replaced by one positioning paragraph.
4. **Presentation.** Sixteen tables became nine, plus three figures: a pipeline diagram, a projected-versus-measured heatmap and a side-by-side of one delegation as stored by each library.
5. **Related work** adds trace-data systems (Canopy, Sifter, Hindsight), log data (Loghub), schema evolution (PRISM), data validation (Deequ), and the current GenAI conventions.
6. **AI Usage Disclosure** added before the references, and E1's code and data are supplied as anonymized supplementary material.

Reviewer 1

1. Results are determined by the projection definitions; validate with traces natively emitted by OTel, OpenInference and OpenLLMetry.

Response. Agreed. The projection study is now presented as a specification-level analysis, and E1 does exactly the suggested validation (Fig. 2). Some predictions hold (no standard planning, memory or guardrail semantics) and some fail in both directions (tool provenance, guardrail kinds, retry loops, delegation in OpenLLMetry).

2. Executable mappings, ambiguous fields, schema versions; OTel+custom 0% kinds but 100% recall.

Response. Sec. IV-B states each anchor and discloses the stale GenAI anchor. Table V lists every attribute accepted as standard when scoring E1, and all library versions are pinned. Sec. III now defines kind coverage as a dedicated kind field, which explains the OTel+custom result.

3. Independent validation is limited; missing-action proxy F1 is near the always-fail baseline.

Response. E1 adds an independent, orchestration-heavy workload with its own ground truth. We agree on the proxy: it does not discriminate between schemas, and it is no longer used as schema evidence (Sec. V-A).

4. Too many tables; add figures.

Response. Done as suggested: a pipeline diagram (Fig. 1) and a schema-by-query heatmap (Fig. 2). The ablation stays a table because its failed-family column carries information a bar chart would lose.

Reviewer 2

Not of interest to the BigData audience.

Response. The revision frames the problem as data quality in a telemetry pipeline: fitness for use of trace stores fed by evolving producer schemas. It also adds a data-validation angle, using recorded replays as regression tests for telemetry pipelines (Sec. VI). We removed the section that argued for the framing.

Reviewer 3

2. *External validity* and *5. projected ≠ implemented*, *9. real telemetry validation*: E1 (Sec. IV-C, V-C).
3. *Deterministic recall, no uncertainty*: E1 scores 1,216 ground-truth instances per condition. Every instance in a family behaved the same, and we report 95% Wilson bounds (Sec. V-C) and instance counts (Fig. 2).
4. *How much AI-generated content*: see the AI Usage Disclosure. Generative AI was used substantively for code, experiments, analysis, figures and text. The authors verified all of it.
5. *Modest scale*: acknowledged under Threats (Scale). The study concerns which semantics reach a store, which does not depend on volume.
6. *Structure does not produce strong diagnosis*: agreed. We removed the diagnostic-proxy and fault-matrix claims and limit the claims to answerability, which is a precondition for diagnosis.
7. *Novelty versus AgentTelemetry*: Sec. II states the boundary. E1's agent, traces, conditions and scoring are new, and the reused fault matrix is removed.
10. *Independent orchestration-heavy dataset; OpenSearch or ClickHouse*: the dataset is provided by E1. A production backend was not feasible in the revision window and is listed as a limitation.
11. *Actual diagnostic performance*: out of the revised scope; see 7.
12. *Storage and query cost*: Sec. IV-B keeps the backend cost figures, and Sec. V-C reports attribute rows per execution for each library.
13. *Content minimization versus privacy*: Sec. VI (Privacy) and E1's findings that two libraries capture content by default and that explicit tool provenance currently requires capturing content.
14. *Evolving standards*: Sec. II gives the current conventions with dates, and related work is expanded.

Reviewer 4

Major 1. Results by construction.

Response. Agreed (see Reviewer 1, point 1). E1 implements the second remedy you proposed: projections replaced by telemetry from actual instrumented executions.

Major 2. Projections may understate conventions (`invoke_agent`, OpenInference `AGENT/CHAIN`).

Response. Correct, and more than we realized: our GenAI anchor was stale (Sec. IV-B). E1 shows what is emitted. OTel GenAI emits `invoke_agent` for the top-level graph but not for the delegated sub-agent. OpenLLMetry emits it for both. OpenInference marks any run whose name contains "agent" as `AGENT`, including a router. No library emits the `plan` or memory operations the conventions now define (Table IX).

Major 3 (a–c) and Q3–Q4. Hallucination 71.4% under structure-only; identical structure-only and full capture; recurring 14.3%.

Response. These came from the prior artifact's controlled matrix. There, detecting a semantic class depended on whether a framework's adapter emitted the spans the detector needed. The detectors read no payload, which is why structure-only and full capture coincide, and 14.3% is one of seven frameworks. That contradicted our text about hallucination and content. These results do not test this paper's question, so we removed them.

Major 3 (d–e). τ-bench F1 does not discriminate; 66.2% action visibility conflates behavior with loss.

Response. Agreed on both. The text now says the proxy does not discriminate and that missing expected actions are actions the agent did not take.

Major 4. Limited Big Data contribution.

Response. Partly addressed through the data-quality framing and the replay-based validation of telemetry pipelines (Sec. VI). Production volume remains a limitation (Sec. VII).

Major 5. Thin increment over AgentTelemetry.

Response. E1 is entirely new, joinability is now measured rather than assessed qualitatively, and the reused fault matrix and proxies are removed.

Minor points.

Response. Kind coverage is defined (Sec. III). Table XIII and its 66.8% were removed. The reference recall being 100% by definition is now stated (Sec. V-A). Related work is expanded. Sections II-B and VI-C are condensed.

Q1. Handling of `invoke_agent/create_agent` in the GenAI view.

Response. They were not mapped; see Major 2.

Q2. Results on traces from the actual instrumentation libraries.

Response. E1, Sec. V-C.